IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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August 26, 2026



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Summary of Methodologies

- Data Collection
- Data Cleaning & Preparation
- Feature Engineering
- Interactive Visualizations
- Model Building & Evaluation

Summary of All Restuls

- Exploratory data analysis results
- Interactive analytics screenshots
- Predictive analytics results

Introduction

Project background and context

In this capstone project, we aim to predict the successful landing of the Falcon 9 first stage. SpaceX advertises rocket launches at a significantly lower cost compared to other providers, largely due to their ability to reuse the first stage of the rocket. By accurately predicting landing success, we can estimate launch costs and provide valuable insights for companies bidding against SpaceX.

Problems you want to find answers

- What factors influence the successful landing of the Falcon 9 first stage?
- How can we accurately predict the landing outcome using machine learning models?
- Which machine learning model performs best in predicting the landing success?

Section 1

Methodology



Methodology

Executive Summary

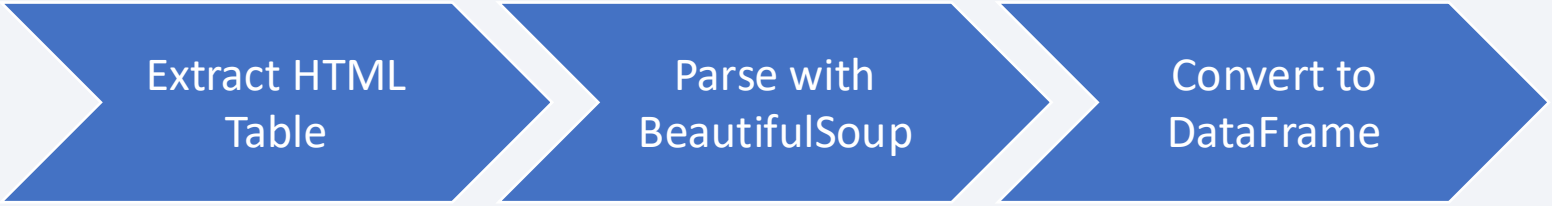
- **Data collection methodology:** Data was sourced from SpaceX API, which provided detailed records of Falcon 9 launches, including launch dates, sites, payloads, and outcomes.
- **Perform data wrangling:** Data cleaning involved handling missing values, standardizing formats, and ensuring consistency. Key features were extracted and new features engineered to enrich the dataset.
- **Perform exploratory data analysis (EDA) using visualization and SQL**
- **Perform interactive visual analytics using Folium and Plotly Dash**
- **Perform predictive analysis using classification models:** Built and evaluated various classification models including Logistic Regression, SVM, KNN, and Decision Trees. Employed GridSearchCV for hyperparameter tuning. Evaluated models based on accuracy, and identified the best performing model for landing success.

Data Collection

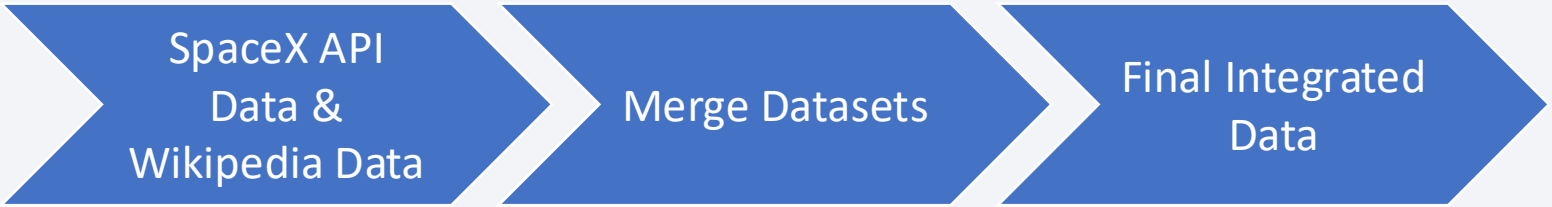
- **Step 1: SpaceX API Request**



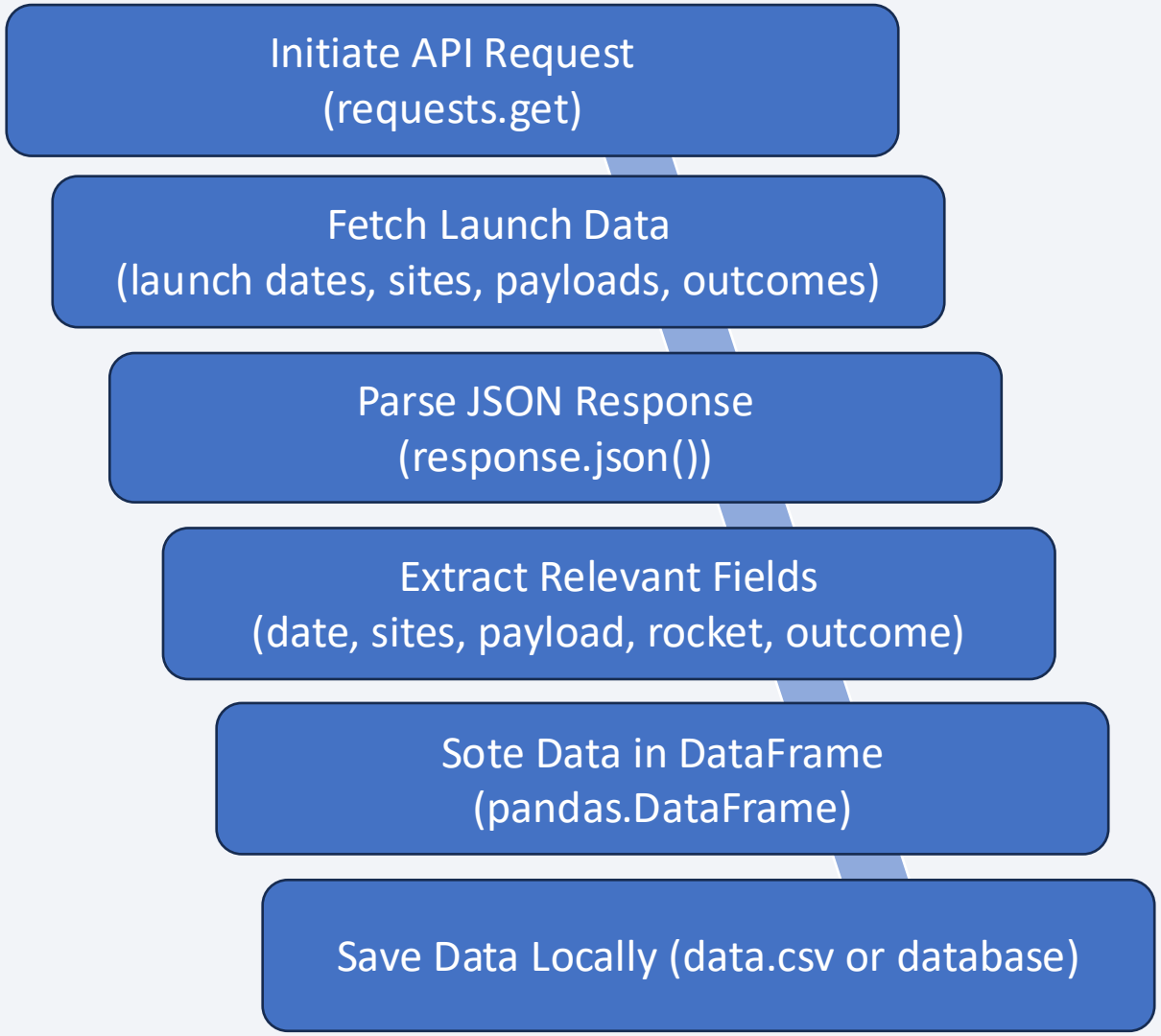
- **Step 2: Web Scraping Wikipedia**



- **Step 3: Data Integration**

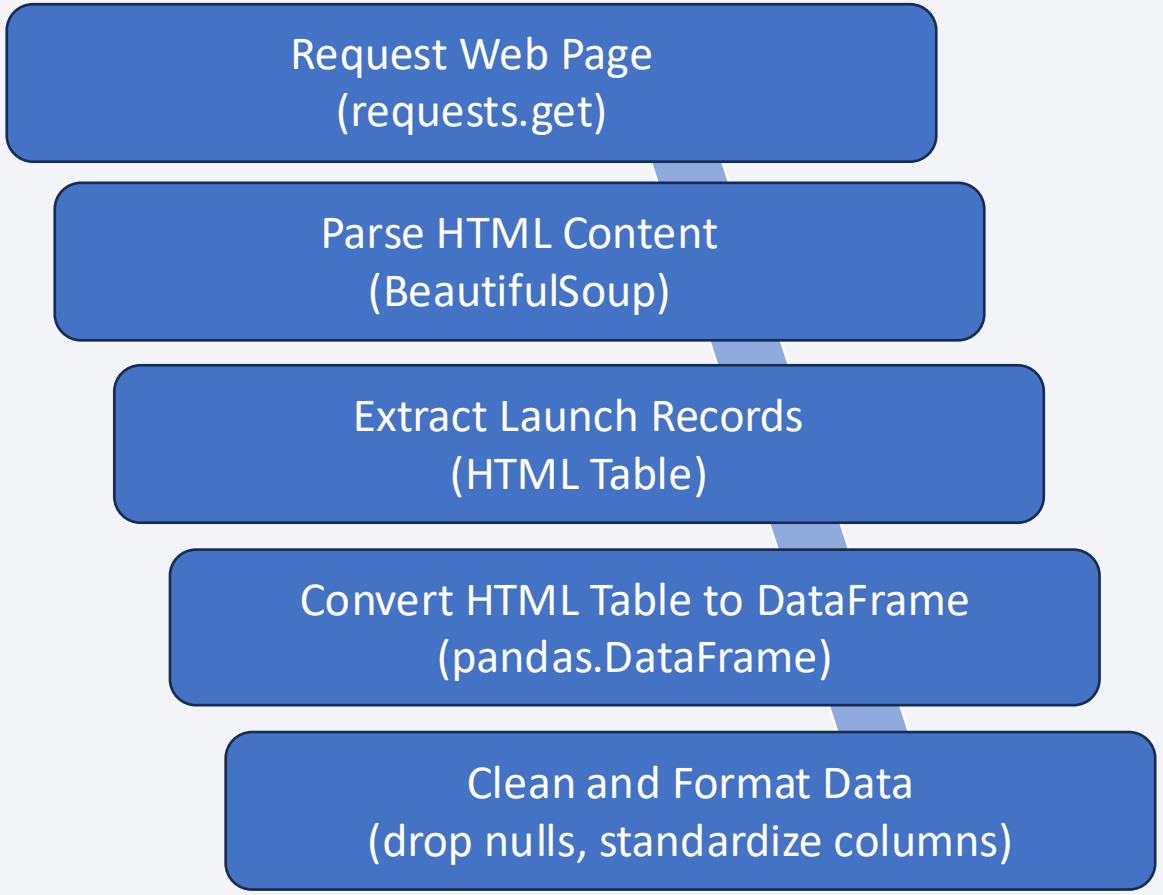


Data Collection – SpaceX API



Github URL: https://github.com/Michelle-Murphy-Green/Applied_Data_Science_Capstone/blob/main/lab1_jupyter-labs-spacex-data-collection-api.ipynb

Data Collection - Scraping

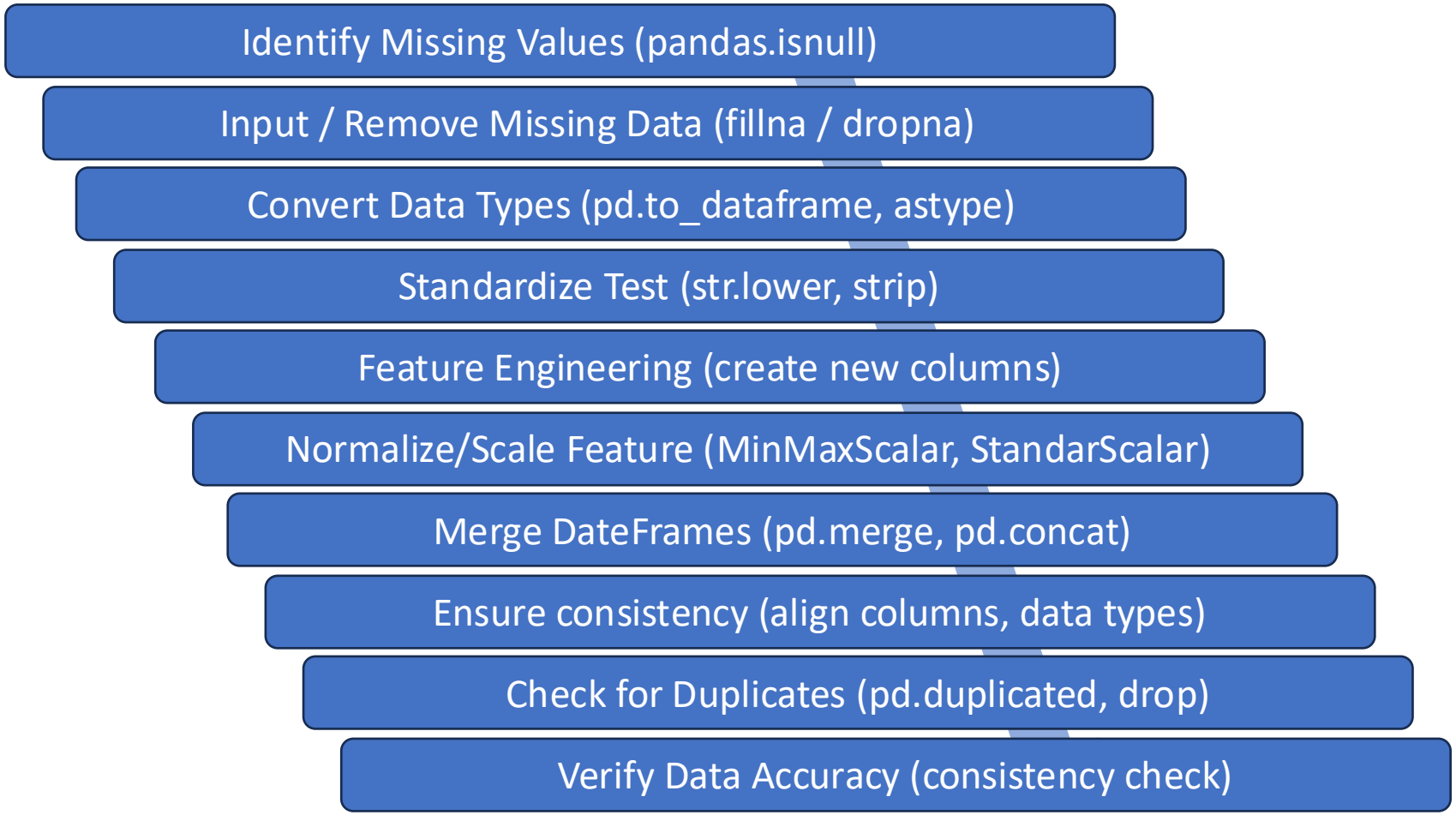


Github URL: https://github.com/Michelle-Murphy-Green/Applied_Data_Science_Capstone/blob/main/lab2_jupyter-labs-webscraping.ipynb

Data Wrangling

- **Overview:** cleaning, transforming, and organizing raw data into a structured format suitable for data analysis
- **Step 1: Data Cleaning** – identifying and filling or removing missing values
- **Step 2: Data Transformation** – converting data types to appropriate formats, standardizing text, creating new features from existing data, and normalizing/scaling numerical features
- **Step 3: Date Integration** – merging datasets collected from different sources into a single dataset
- **Step 4: Data Validation** – checking for and removing duplicate records and verifying the accuracy and consistency of data entries

Data Wrangling



Github URL: https://github.com/Michelle-Murphy-Green/Applied_Data_Science_Capstone/blob/main/lab3_labs-jupyter-spacex-Data%20wrangling.ipynb

EDA with Data Visualization

- **Overview:** visually exploring and summarizing the main characteristics of a dataset with the goal to understand the data's distribution, identify patterns, and uncover relationships between variables
- **Histograms:** visualize the distribution of numerical variables to understand the spread and central tendency of the data, identify outliers, and skewness
- **Bar Charts:** compare categorical variables to compare frequencies or proportions and highlight patterns
- **Line Charts:** track trends over time to reveal temporal patterns and performance trends or changes over time

EDA with Data Visualization

- **Scatter Plots:** explore relationships between two numerical variables to identify correlations or dependencies between variables
- **Heatmaps:** visualize correlation matrices between multiple numerical variables to identify strong correlations between variables (pos or neg)
- **Box Plots:** display distribution of numerical data through their quartiles to determine spread of data, identify outliers, and compare distributions

Github URL: https://github.com/Michelle-Murphy-Green/Applied_Data_Science_Capstone/blob/main/lab4_edadataviz.ipynb

EDA with SQL

- **Aggregate Queries:**
  - Calculated total number of launches
  - Counted successful and failed launches
  - Calculated success rates by launch site and rocket type
- **Join Queries:**
  - Joined tables to link launch records with additional details
  - Combined datasets for comprehensive analysis
- **Filtering Queries**
  - Filtered data to focus on specific launch outcomes
  - Applied conditions to extract launches based on criteria such as launch data or rocket configuration
- **Sorting Queries**
  - sorted data to identify trends for outliers
  - Ordered launches by date or success rate

Github URL: https://github.com/Michelle-Murphy-Green/Applied_Data_Science_Capstone/blob/main/lab5_jupyter-labs-eda-sql-coursera_sqllite.ipynb

Build an Interactive Map with Folium

- **Markers**
  - Used to indicate launch sites with each marker representing a specific geographical location where SpaceX launches have occurred
  - Reason: spatial reference to help users identify where SpaceX has conducted launches geographically
- **Circles**
  - Used to visually represent proximity zones around launch sites to identify areas that may be effected
  - Reason: illustrates potential impact zones and represents safety perimeters or operational boundaries
- **Lines**
  - Used to connect launch sites to other relevant locations and points of interest
  - Reason: enhances understanding of spatial relationships and dependences between launch sites and other relevant features or locations in the area

Github URL: https://github.com/Michelle-Murphy-Green/Applied_Data_Science_Capstone/blob/main/lab6_lab_jupyter_launch_site_location.ipynb

Build a Dashboard with Plotly Dash

- **Plots/Graphs Added:**
  - Success Pie Chart
    - Displays the distribution of successful and failed launches to visualize the overall success rate and trends
    - **Reason**: provides a quick overview of mission success rates
  - Succes Payload Scatter Plot
    - Shows the relationship between payload mass and launch success and how payload mass influences mission outcomes
    - **Reason**: helps identify correlations between payload characteristics and launch outcomes

- **Interactions Added:**
  - Launch Site Dropdown
    - Enables users to select specific launch sites for focused analysis based on geographic locations
    - **Reason**: helps focus analysis and allows for regional insights and comparisons across launch sites
  - Range Slider for Payload
    - Allows users to adjust payload mass ranges dynamically
    - **Reason**: enables detailed analysis and insights into payload-related performance factors

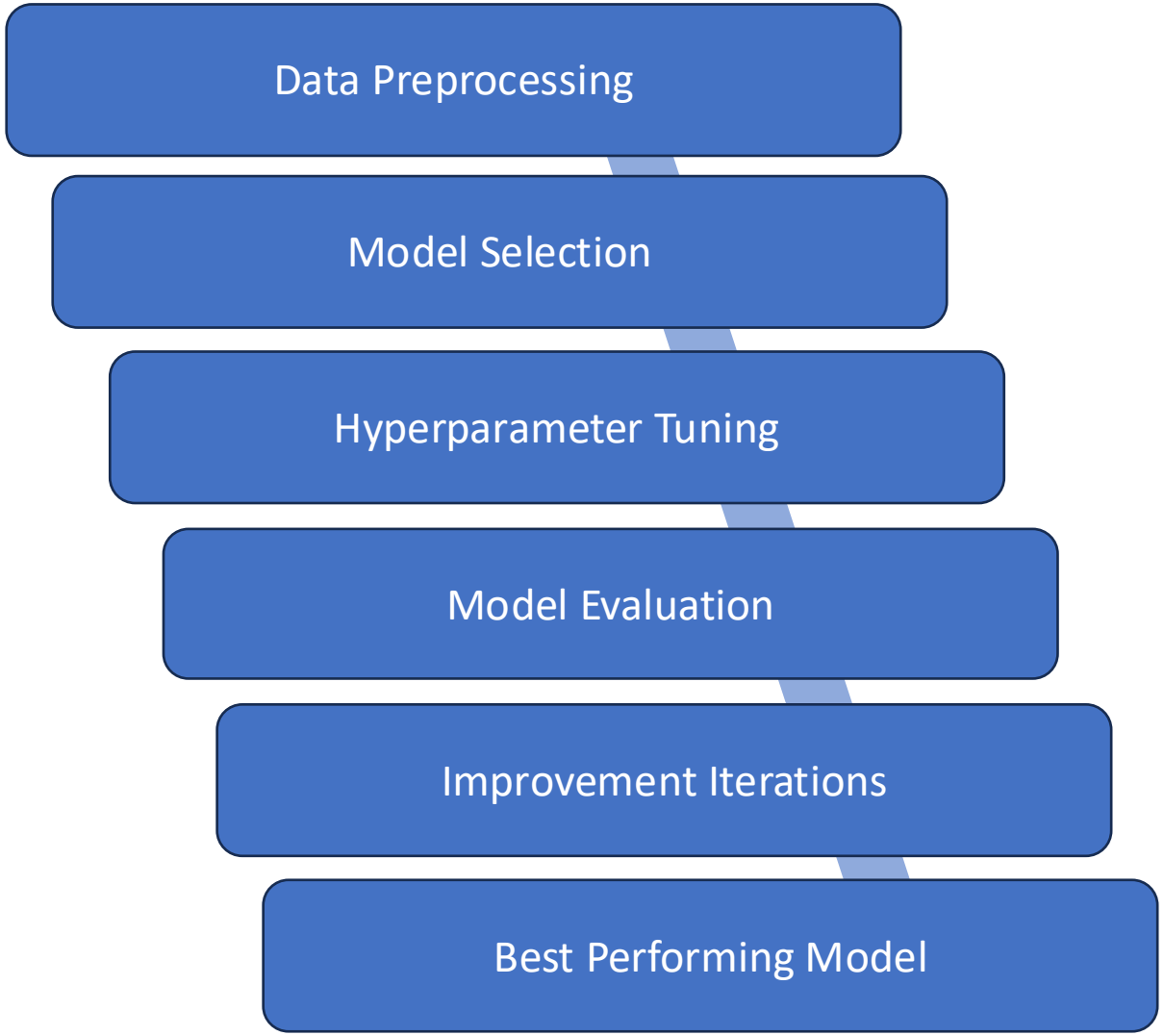
Github URL: https://github.com/Michelle-Murphy-Green/Applied_Data_Science_Capstone/blob/main/lab7_spacex-dash-app.py

Predictive Analysis (Classification)

- 1 - Data Preprocessing
  - Standardized features to ensure all variable contribute equally
  - Split data into training and test sets for model validation
- 2 - Model Selection
  - Explored multiple classification algorithms: SVM, Decision Trees, K-Nearest Neighbors
  - Chose algorithms suitable for binary classification tasks based on project requirements
- 3 - Hyperparameter Tuning
  - Used GridSearchCV to systematically search for optimal hyperparameters
  - Tuned parameters such as C (SVM), max_depth (Decision Trees), and n_neighbors (K-Nearest Neighbors)
- 4 - Model Evaluation
  - Evaluated models using cross-validation techniques to ensure robustness and generalizability
  - Utilized metrics like accuracy, precision, recall, and F1-score to asses model performance
- 5 - Improvement Iterations
  - Explored multiple classification algorithms: SVM, Decision Trees, K-Nearest Neighbors
  - Fine-tuned hyperparameters to maximize predictive accuracy and reliability
- 6 - Selection of Best Performing Model
  - Identified the model with the highest accuracy on the test set as the best performer
  - Considered training and test set performance to avoid overfitting and ensure real-world application

Github URL: https://github.com/Michelle-Murphy-Green/Applied_Data_Science_Capstone/blob/main/lab8_SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

Predictive Analysis (Flowchart)

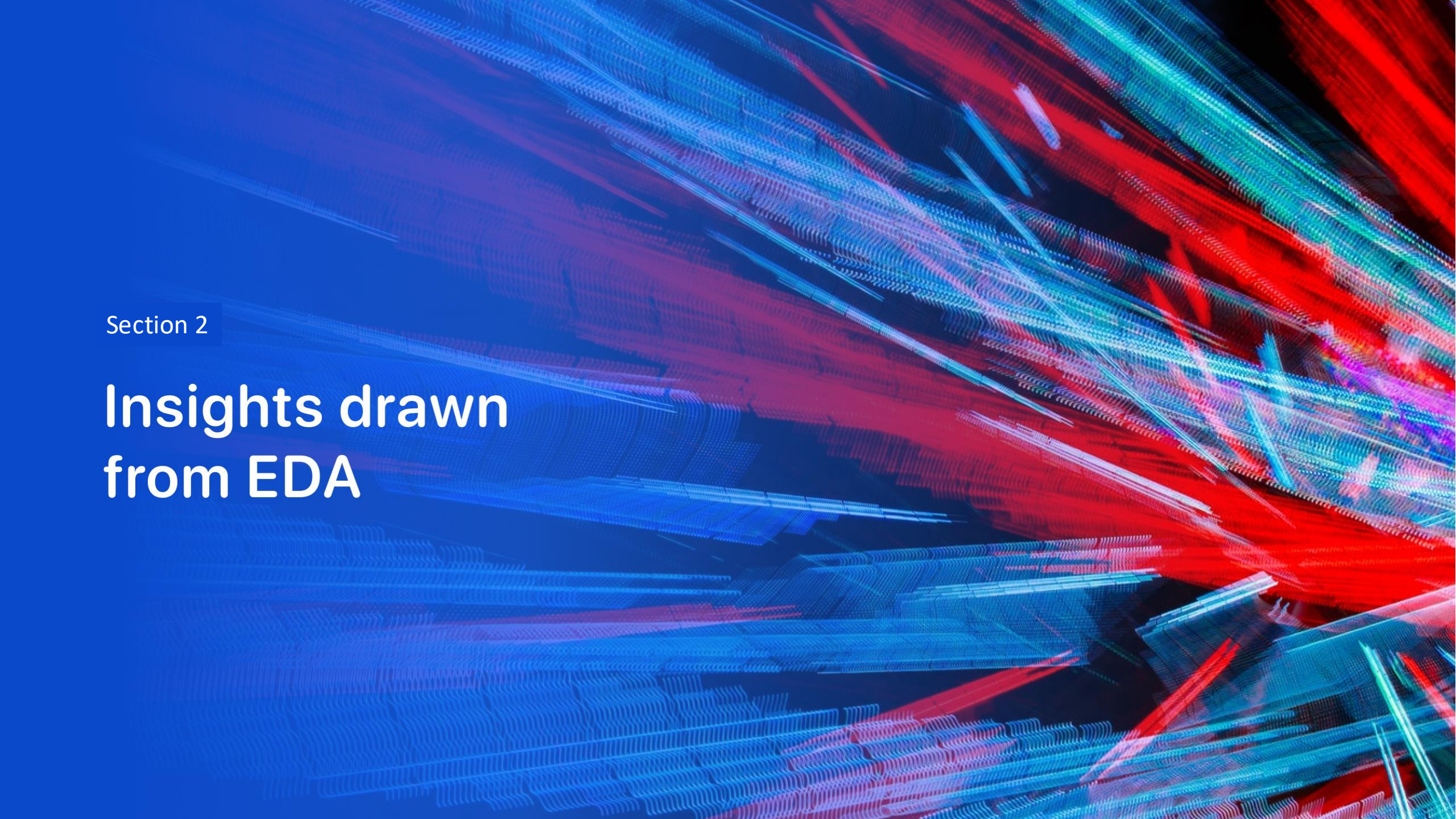


Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

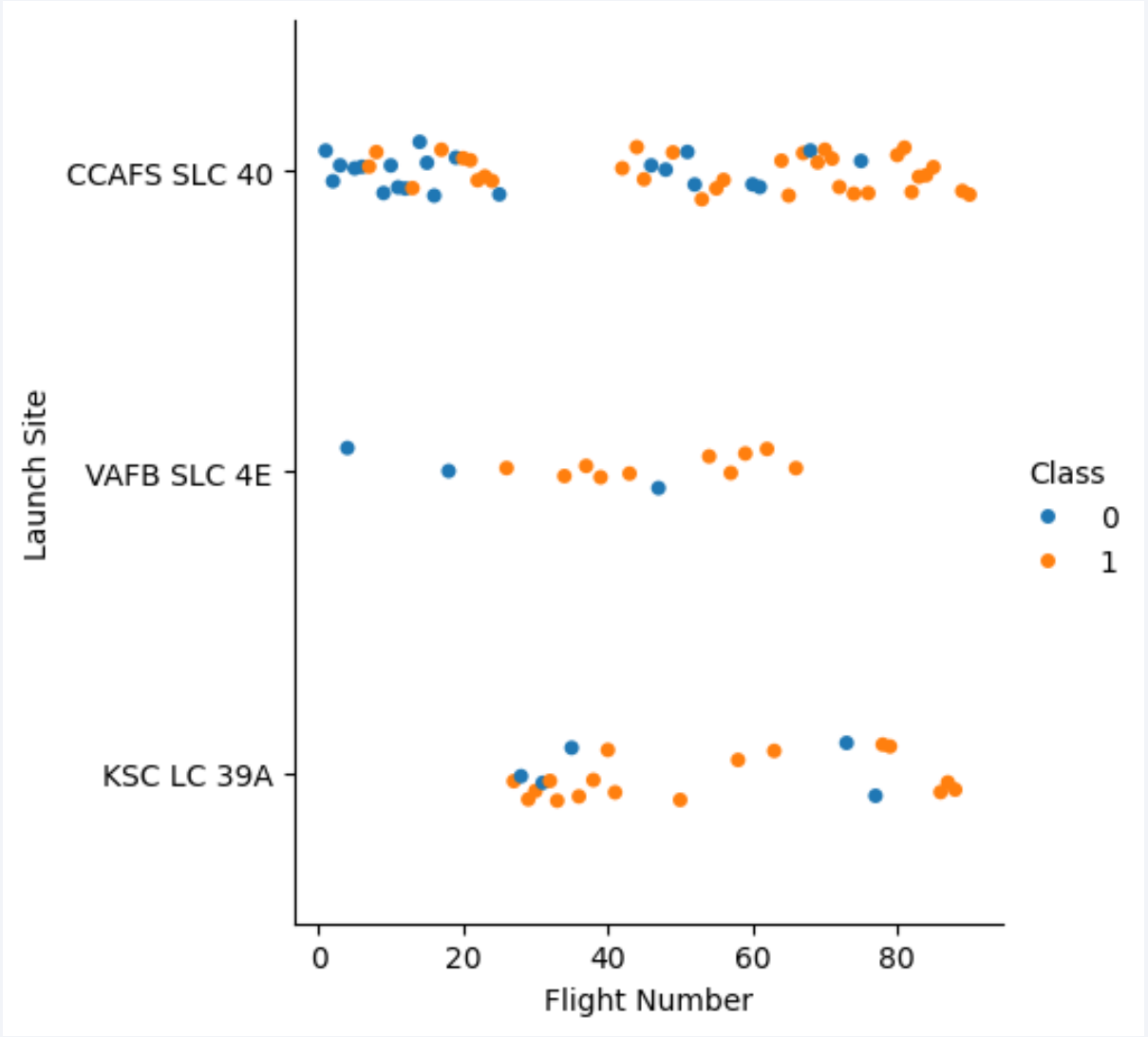
Section 2

Insights drawn from EDA



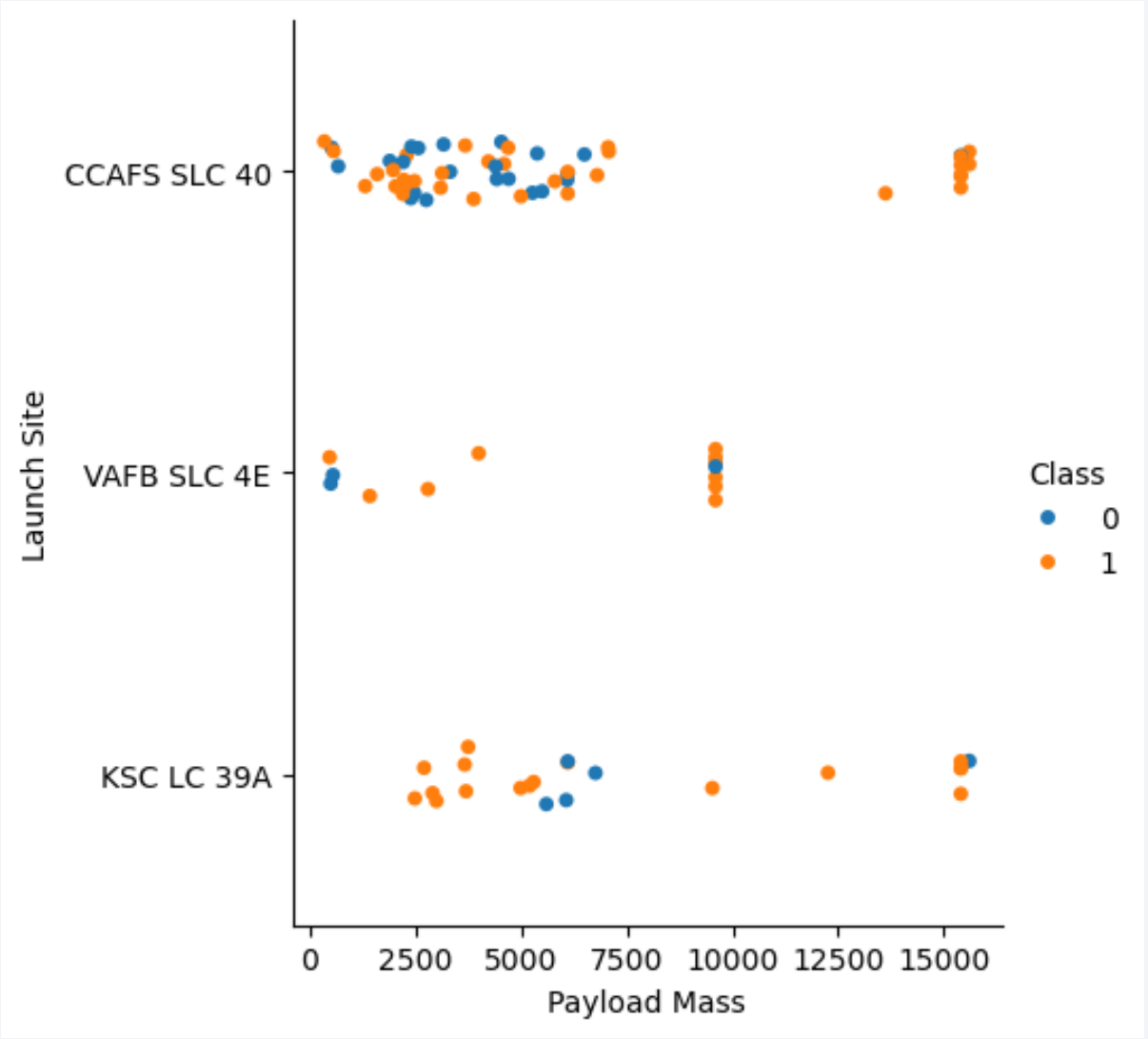
Flight Number vs. Launch Site

- **Mixed Outcomes At Major Launch Sites:** Both CCAFS SLC40 and KSC LC 39A have a mix of successful (orange) and unsuccessful (blue) landings, indicating that factors other than the launch site itself may influence the landing success
- **Consistent Activity Across Flight Numbers:** Launches are spread across a wide range of flight numbers at all sites, suggesting consistent activity over time without a clear trend of increasing or decreasing landing success



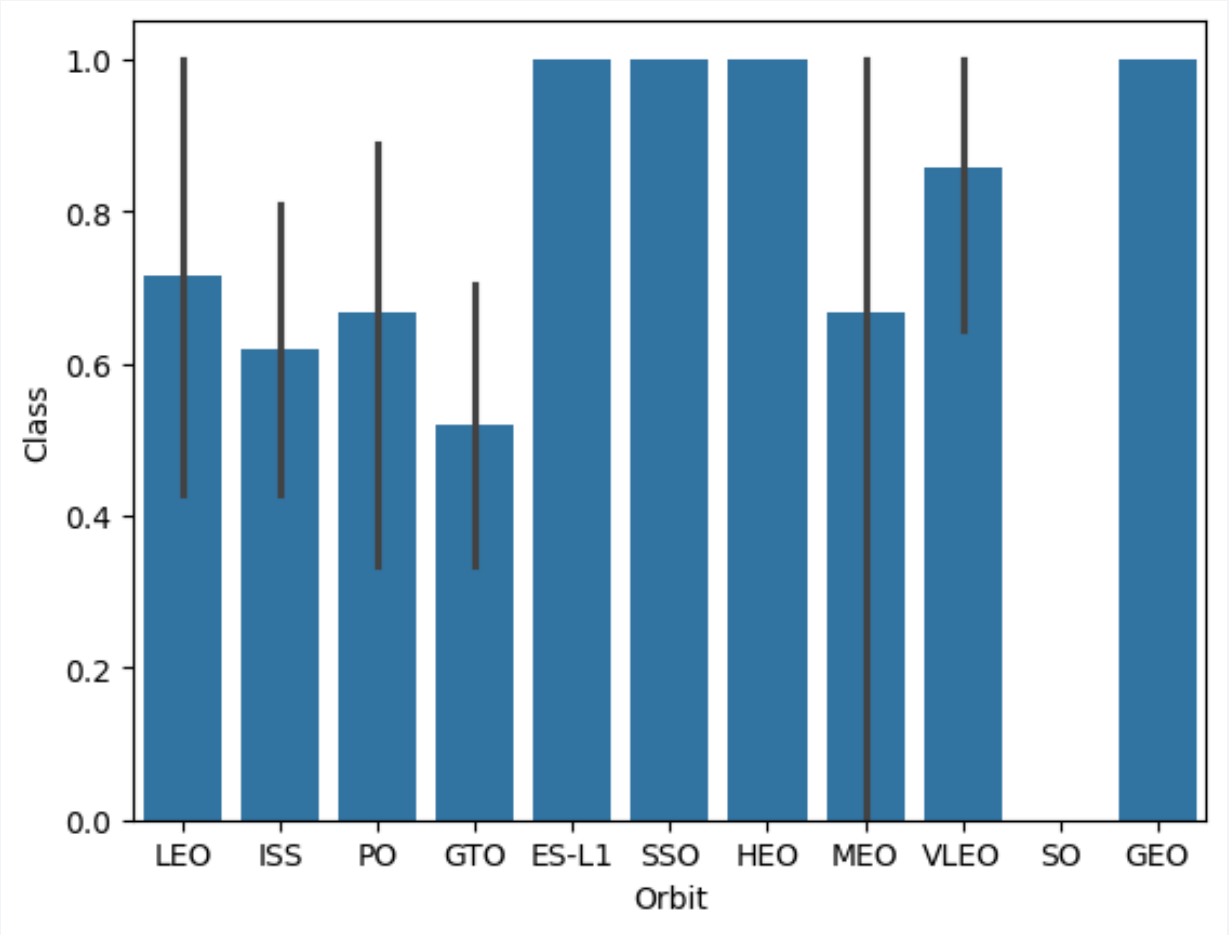
Payload vs. Launch Site

- **Payload Distribution:** Most launches from the CCAFS SLC 40 sites handle payloads below 10,000 kg, while the VAFB SLC 4E and KSC LC 39A sites have a wider range of payload masses, indicating varied mission profiles
- **High-Capacity Launches:** The KSC LC 39A sites is frequently used for launching heavier payloads, with multiple launches carrying over 15,000 kg, suggesting its suitability for high-capacity missions



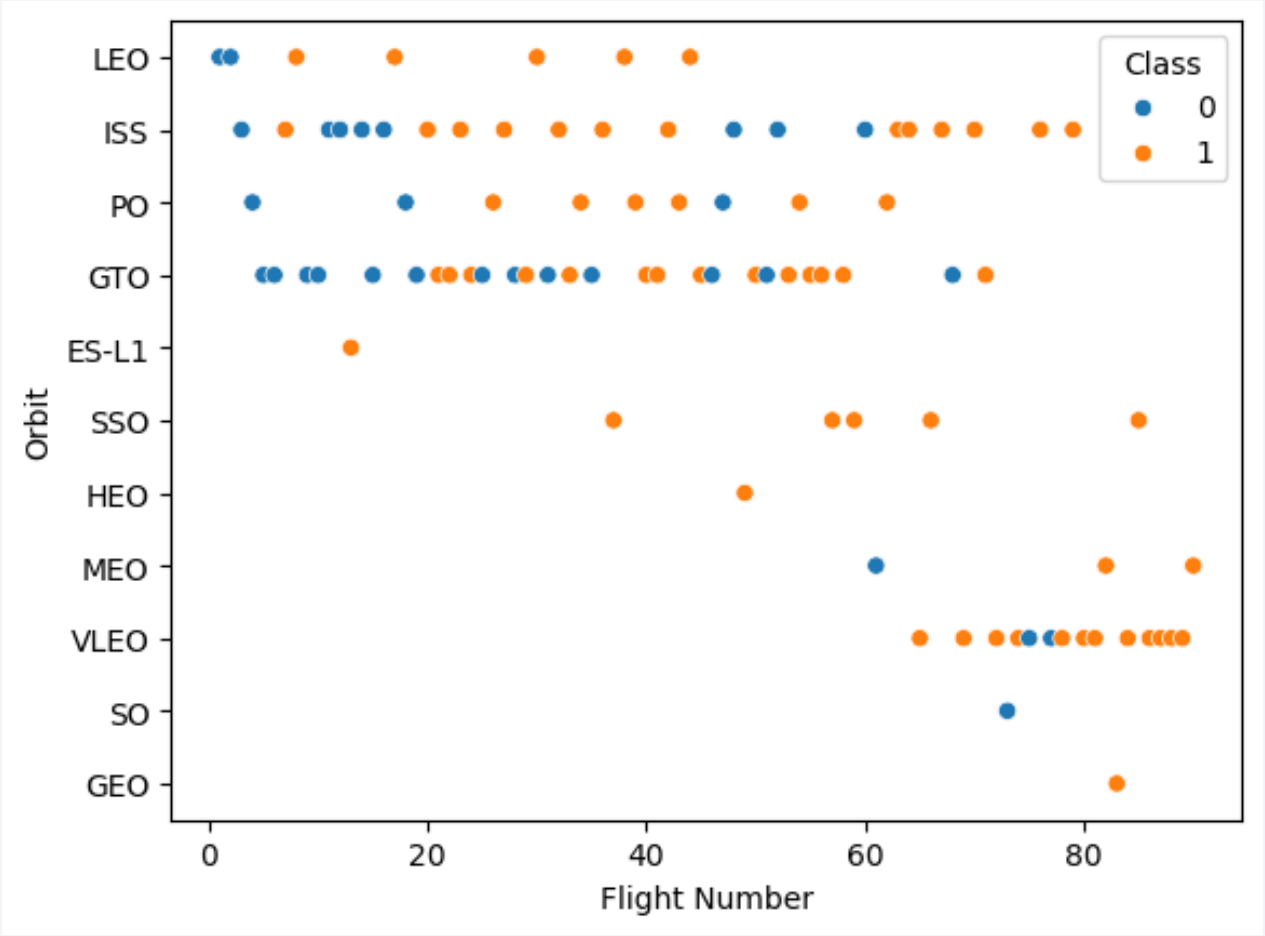
Success Rate vs. Orbit Type

- **High Success Rates:** Missions to VLEO, ES-L1, HEO, and SSO orbits have achieved a perfect success rate, indicating these orbits are highly reliable for successful first stage landings
- **Lower Success Rates for GTO:** The GTO orbit types shows a significantly lower success rate compared to other orbit types, suggesting that missions to this orbit may involved greater challenges or complexities



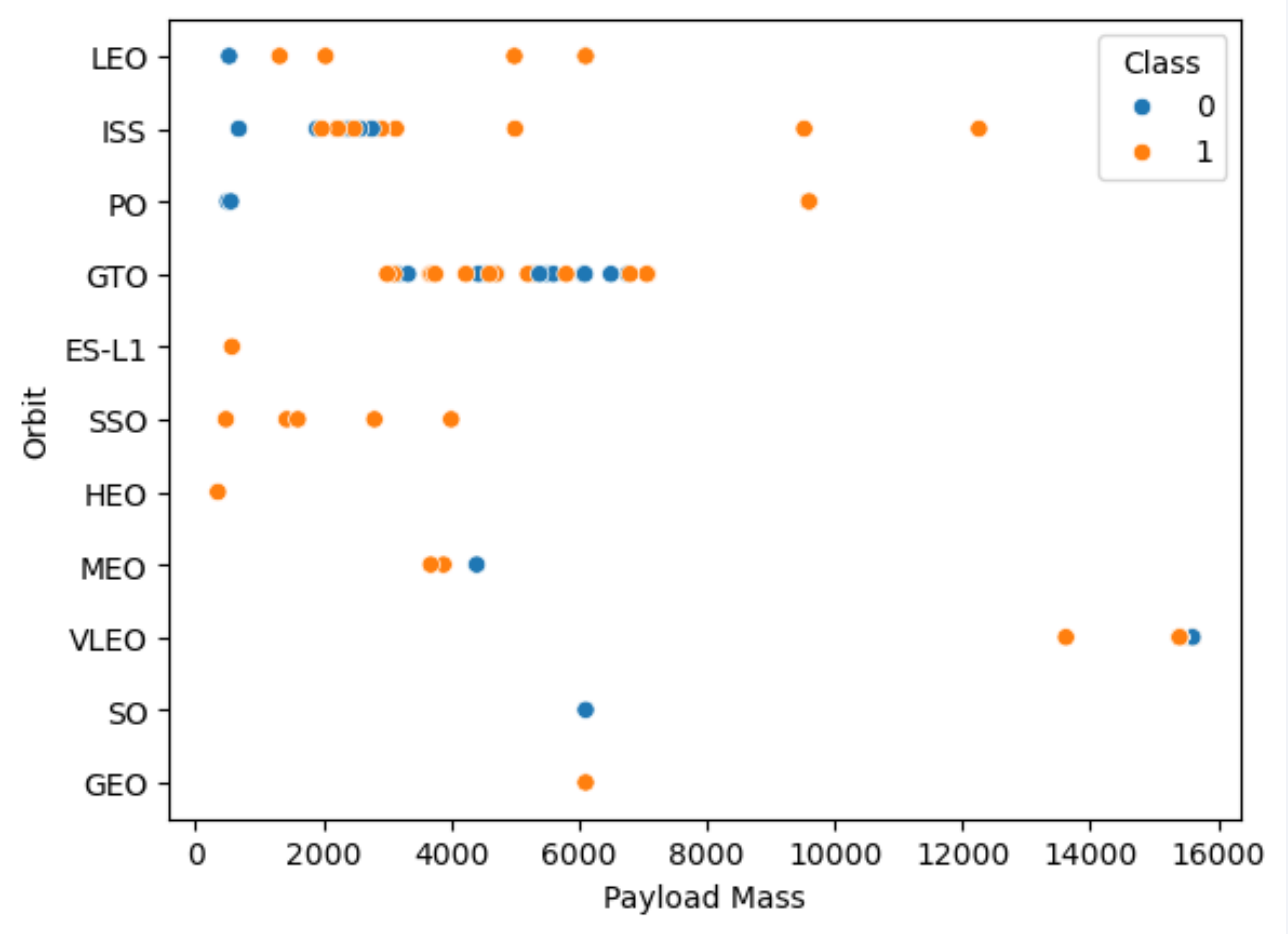
Flight Number vs. Orbit Type

- **Increased Success Over Time:** The success rates of Falcon 9 launches improves significantly with higher flight numbers, indicating that experience and iterative improvements contribute to better outcomes
- **Orbit-Specific Performance:** Early flights to GOT and ISS orbits had mixed outcomes, but recent missions to these orbits show a higher success rate, reflecting advancements in mission planning and execution



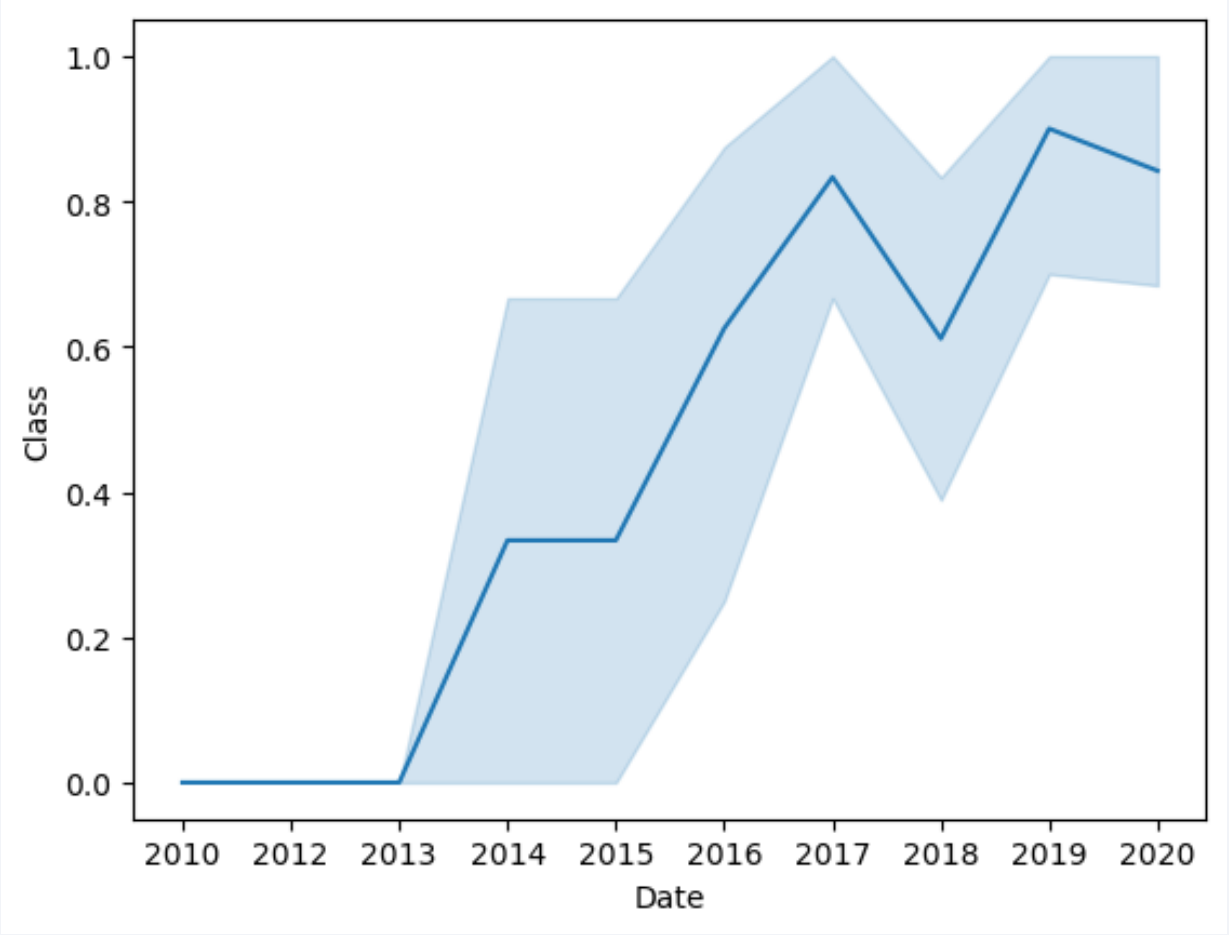
Payload vs. Orbit Type

- Successful landings are more frequent across all orbit types, especially for payloads less than 6,000 kg
- Higher payload masses (above 10,000 kg) show a mix of success and failures, indicating increased difficulty with heavier payloads.



Launch Success Yearly Trend

- The annual launch success rate has shown a significant improvement from 2013 onwards, reaching over 80% by 2020
- Despite a dip in 2018, the overall trend indicates increasing reliability and success in Falcon 9 launches over the years



All Launch Site Names

```
%sql SELECT DISTINCT "Launch_Site" FROM SPACEXTABLE;
```

```
 * sqlite:///my_data1.db
Done.
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

```
%sql SELECT * FROM SPACEXTABLE WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5;
```

```
 * sqlite:///my_data1.db
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

```
%sql SELECT SUM("PAYLOAD_MASS__KG_") FROM SPACEXTABLE WHERE "Customer" LIKE 'NASA (CRS)';
```

```
 * sqlite:///my_data1.db
Done.
```

SUM("PAYLOAD_MASS__KG_")
45596

Average Payload Mass by F9 v1.1

```
%sql SELECT AVG("PAYLOAD_MASS__KG_") FROM SPACEXTABLE WHERE "Booster_Version" LIKE 'F9 v1.1%';
```

```
 * sqlite:///my_data1.db
Done.
```

AVG("PAYLOAD_MASS__KG_")
2534.6666666666665

First Successful Ground Landing Date

```
%sql SELECT MIN("Date") FROM SPACEXTABLE WHERE "Landing_Outcome" LIKE 'Success (ground pad)'
```

```
 * sqlite:///my_data1.db
Done.
```

MIN("Date")
2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

```
%sql SELECT "Booster_Version" FROM SPACEXTABLE WHERE "Landing_Outcome" LIKE 'Success (drone ship)' AND "PAYLOAD_MASS__KG_" BETWEEN 4000 AND 6000
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

```
%%sql SELECT mission_outcome, COUNT(*) as COUNT
FROM SPACEXTBL
GROUP by mission_outcome ORDER BY mission_outcome;
```

```
 * sqlite:///my_data1.db
Done.
```

Mission_Outcome	COUNT
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

```
%sql SELECT "Booster_Version" FROM SPACEXTABLE WHERE "PAYLOAD_MASS__KG_" = (SELECT MAX("PAYLOAD_MASS__KG_") FROM SPACEXTABLE)
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

```
%sql SELECT substr(Date, 6, 2) AS Month, landing_outcome, booster_version, launch_site FROM SPACEXTBL WHERE substr(Date, 1, 4) = '2015';
```

```
 * sqlite:///my_data1.db
Done.
```

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
02	Controlled (ocean)	F9 v1.1 B1013	CCAFS LC-40
03	No attempt	F9 v1.1 B1014	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40
04	No attempt	F9 v1.1 B1016	CCAFS LC-40
06	Precluded (drone ship)	F9 v1.1 B1018	CCAFS LC-40
12	Success (ground pad)	F9 FT B1019	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
%sql SELECT "Landing_Outcome", COUNT(*) AS COUNT FROM SPACEXTABLE WHERE "Date" Between '2010-06-04' AND '2017-03-20' GROUP BY "Landing_Outcome" ORDER BY "(
```

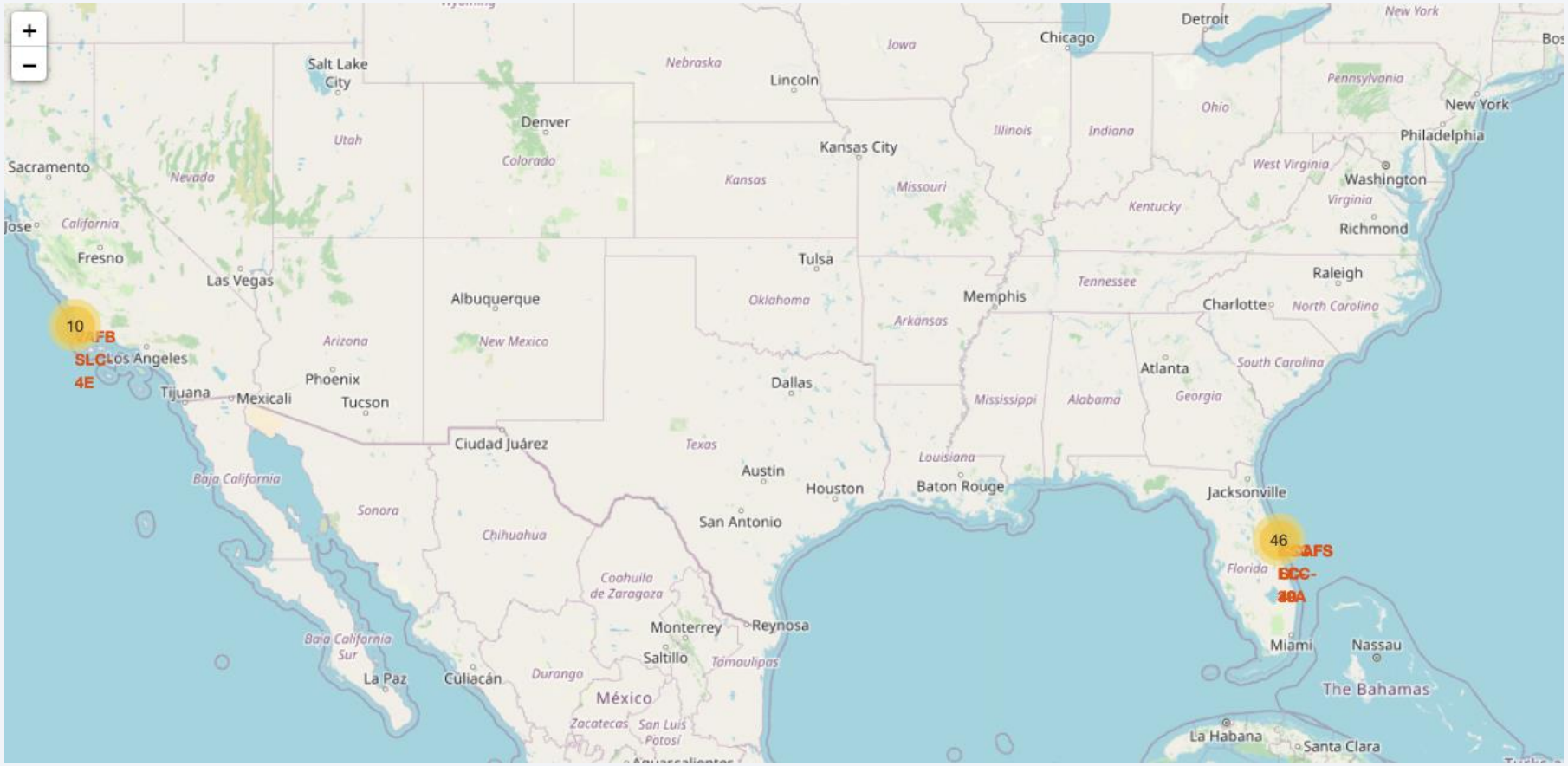
```
 * sqlite:///my_data1.db
Done.
```

Landing_Outcome	COUNT
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1


Section 3
Launch Sites
Proximities Analysis

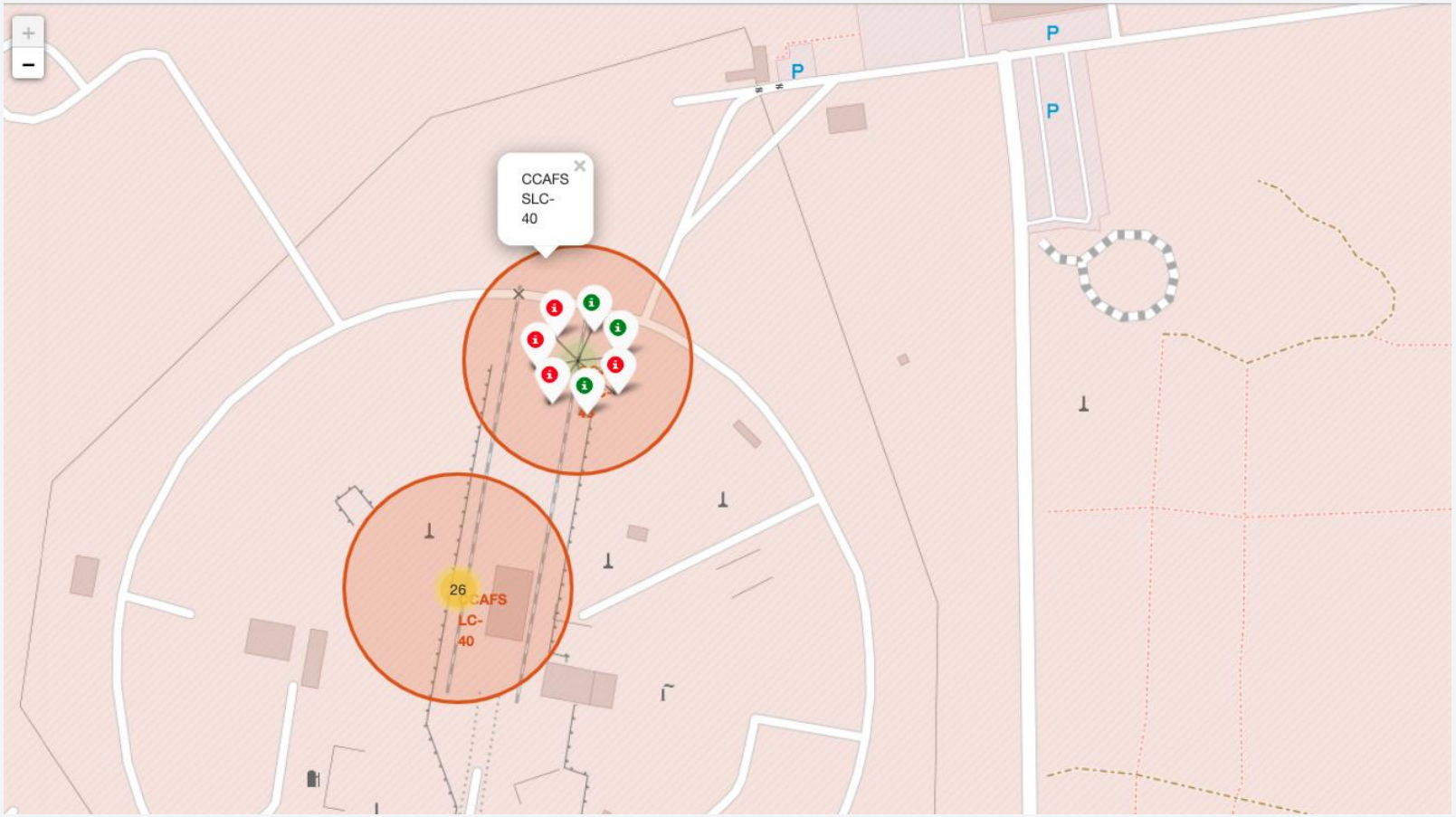
Marking All Launch Sites

- The launch sites are all somewhat close to the equator and close to the coast
- There have been 10 launches in California and 46 launches in Florida



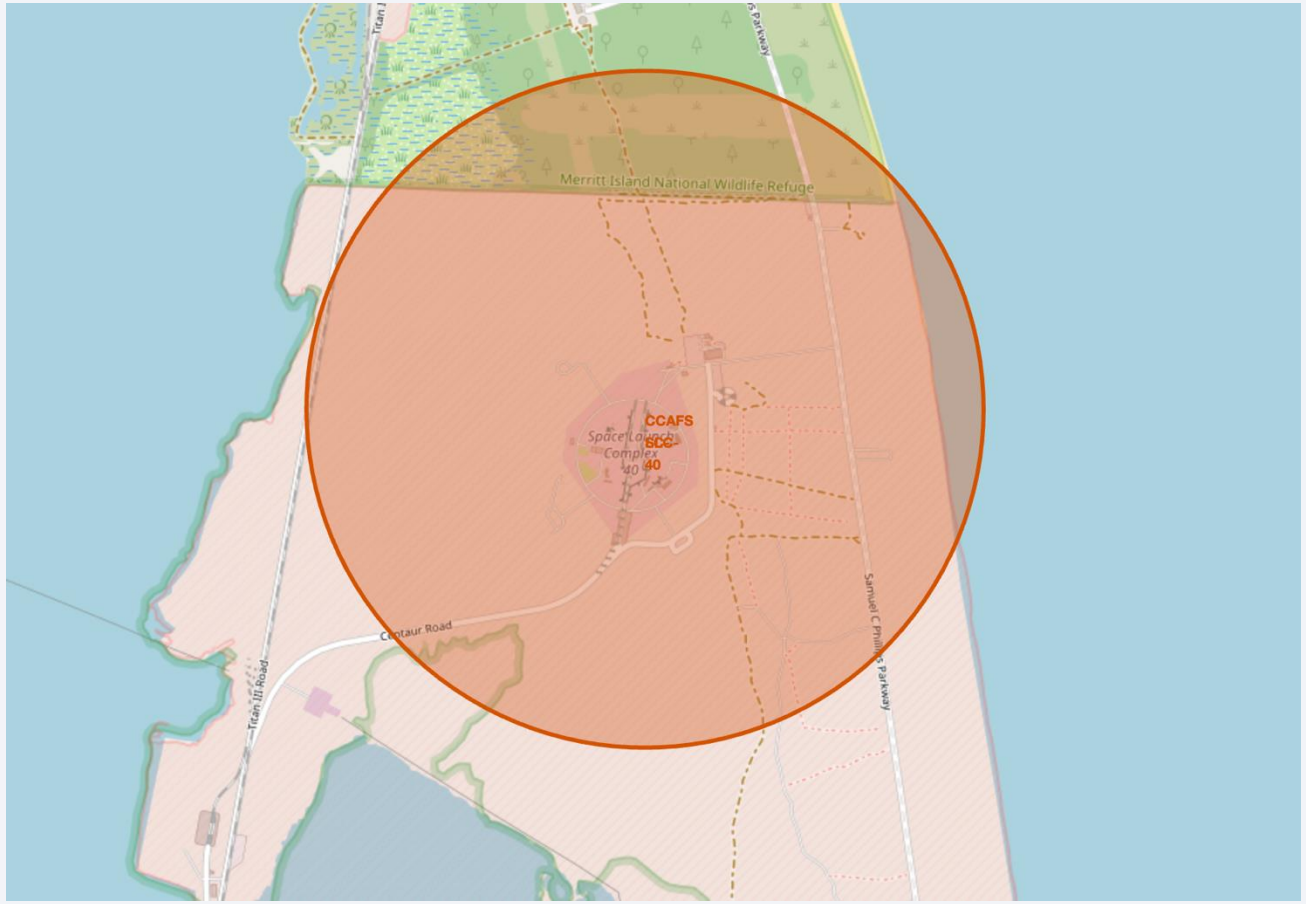
Marking Successful/Failed Launches

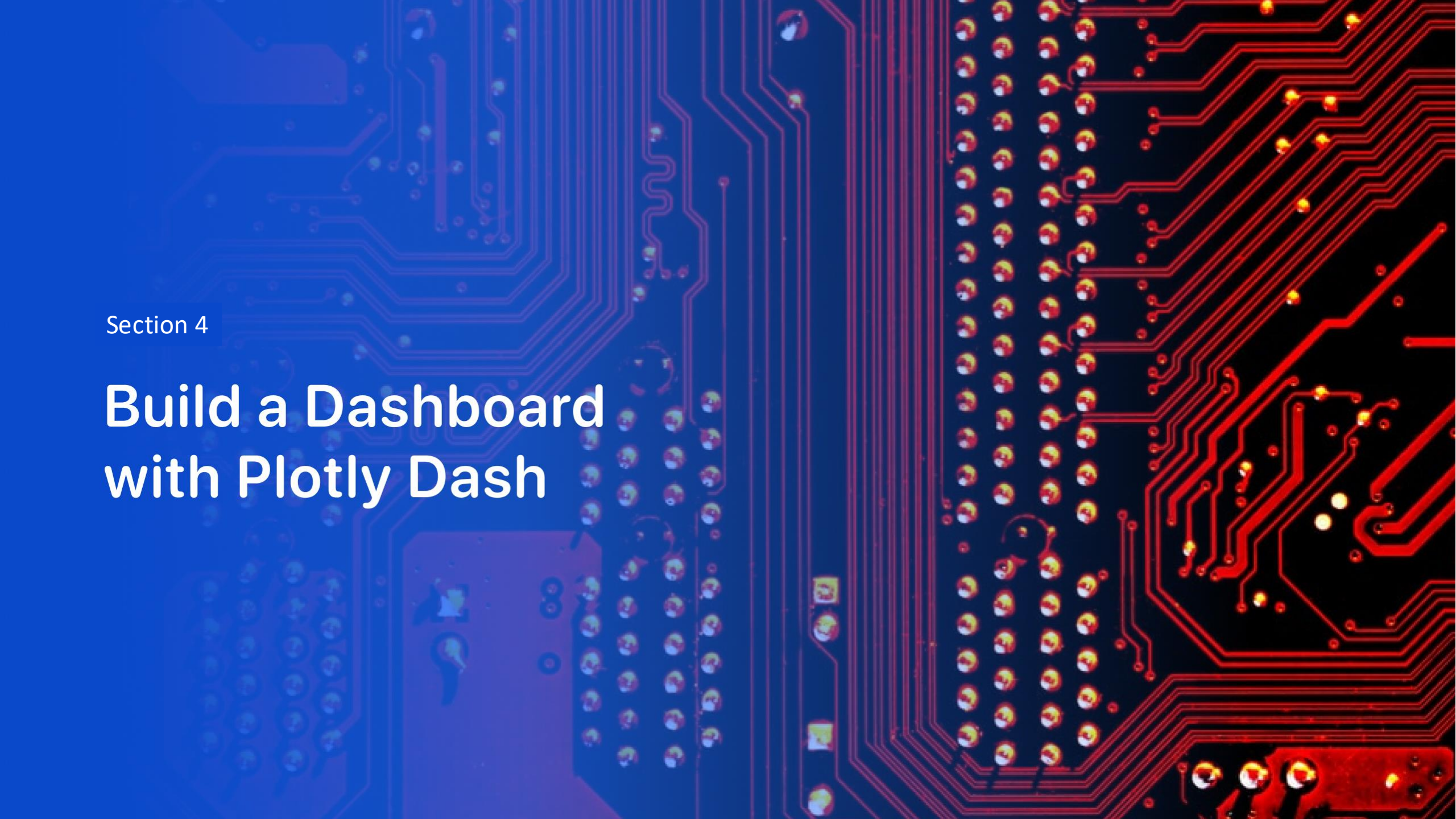
- Successful launches are marked in green, unsuccessful in red
- CCAFS SCL-40 has a low success rate



Distances Between Launches Sites & Proximities

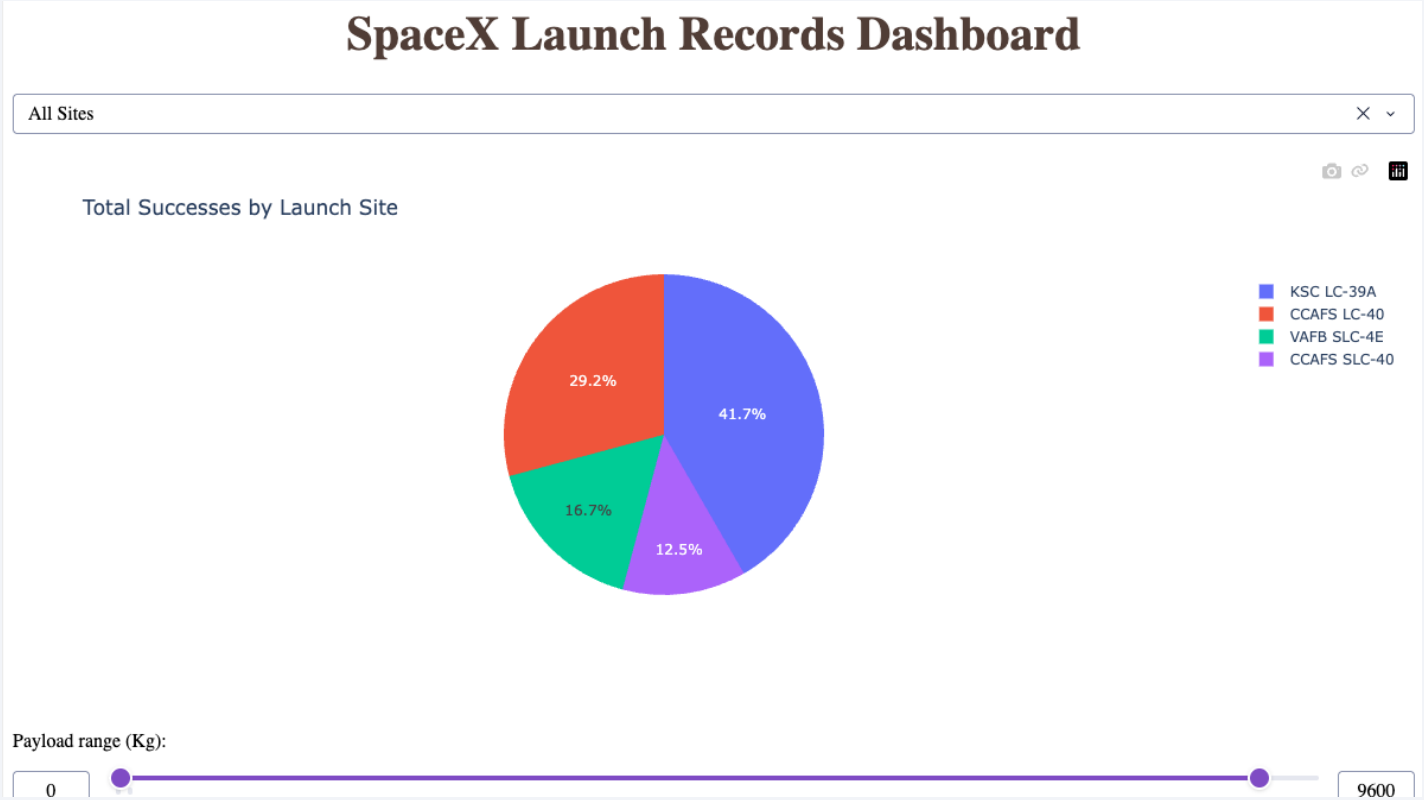
- The CCAFS SLC-40 launch site is close to both the coastline, a highway, and a railway
- Coastline: 54 km
- Highway: 20 km
- Railway: 24 km




Section 4
Build a Dashboard with Plotly Dash

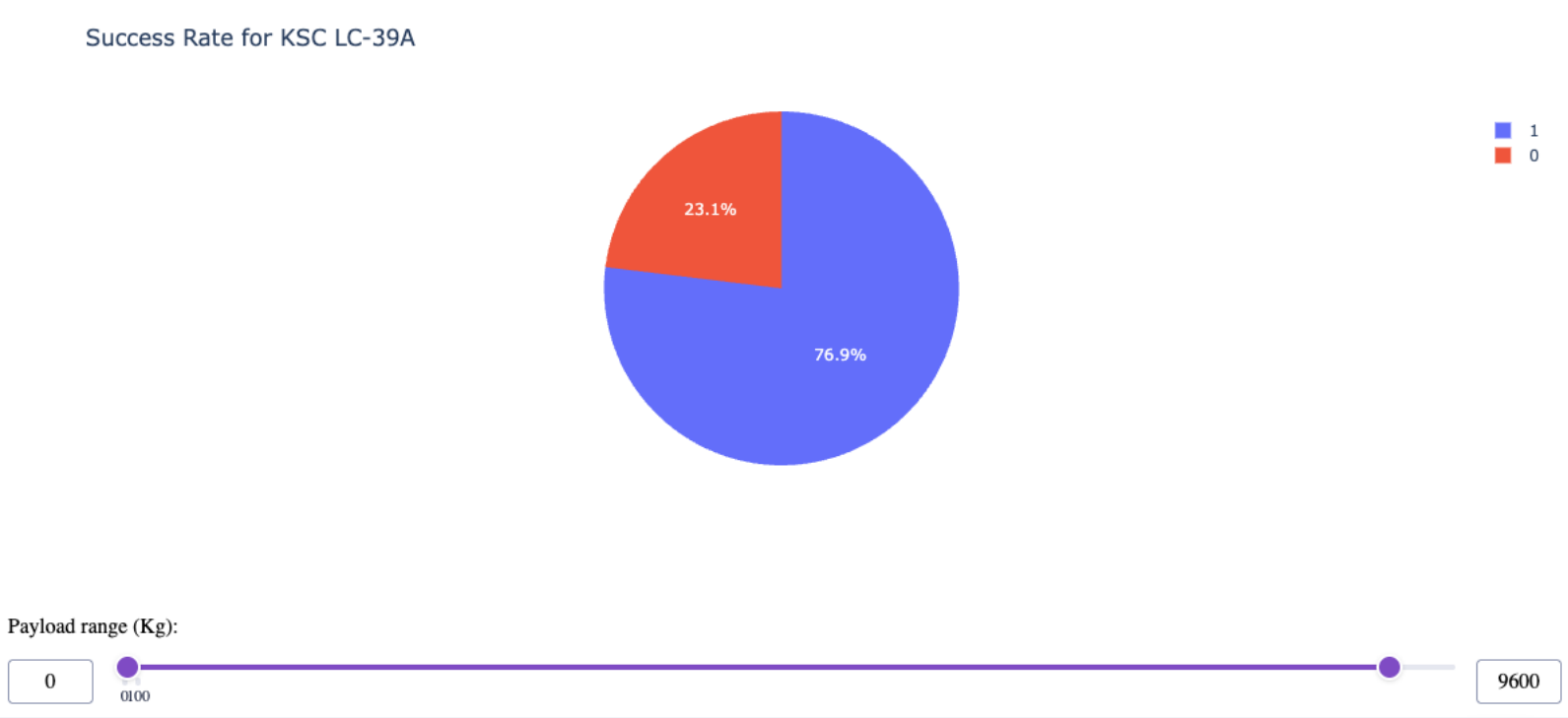
Launch Success Count for All Sites

- The KSC LC-39A launch site has the highest number of successful launches, making up 41.7% of the total successes
- The CCAFS SLC-40 sites has the lowest number of successful launches



Launch Sites with Highest Launch Success Ratio

- The KSC LC-39A launch site has had 76.9% successful launches and 23.1% unsuccessful launches
- The high success rate shows this site's effectiveness and reliability



Key Insights from SpaceX Launch Data Dashboard

- A payload mass between 2000 and 4000 kg has the most success rate
- The booster version ‘FT’ is the most frequently used with the highest success rate across all payload masses



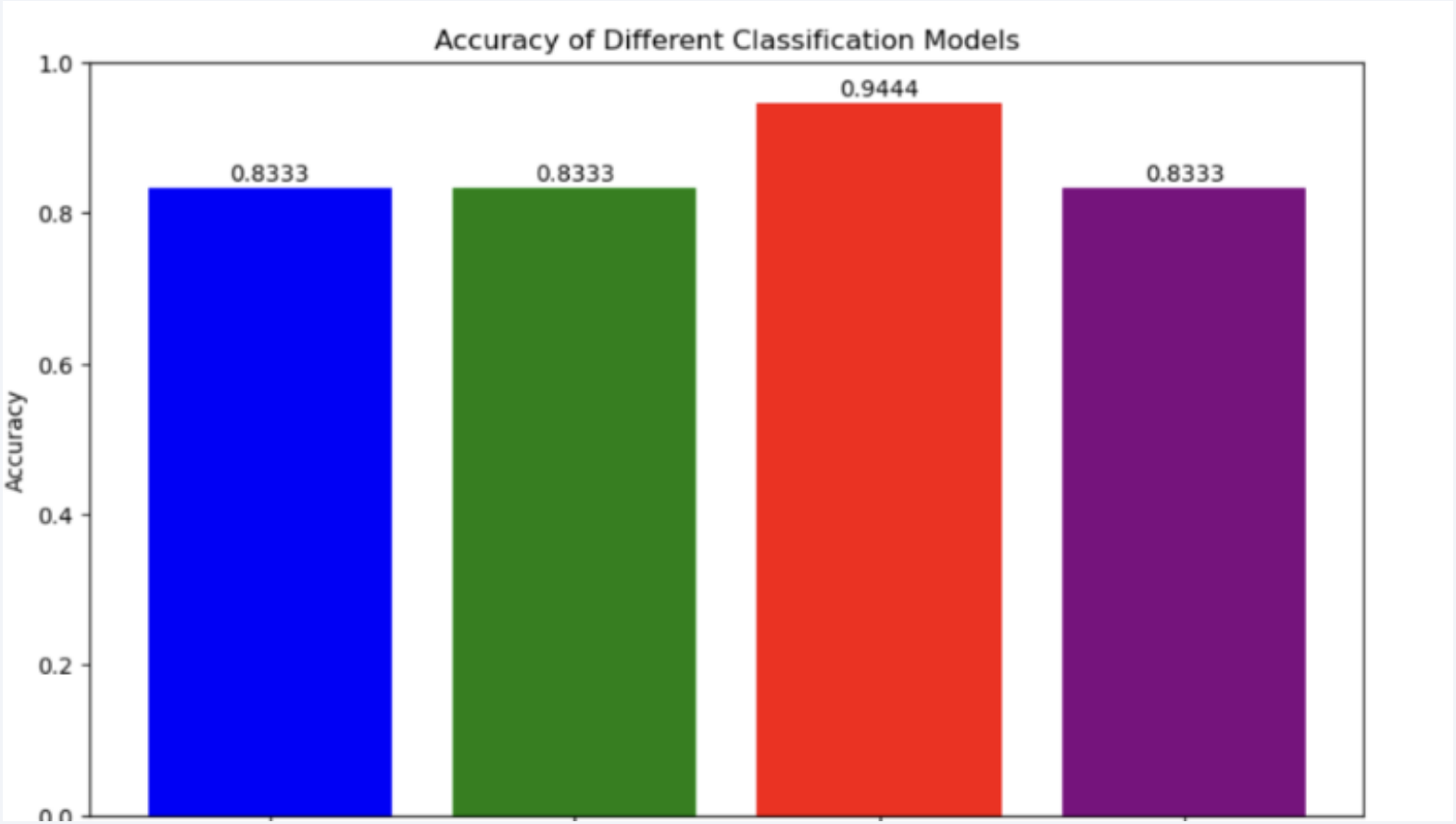
Section 5

Predictive Analysis (Classification)



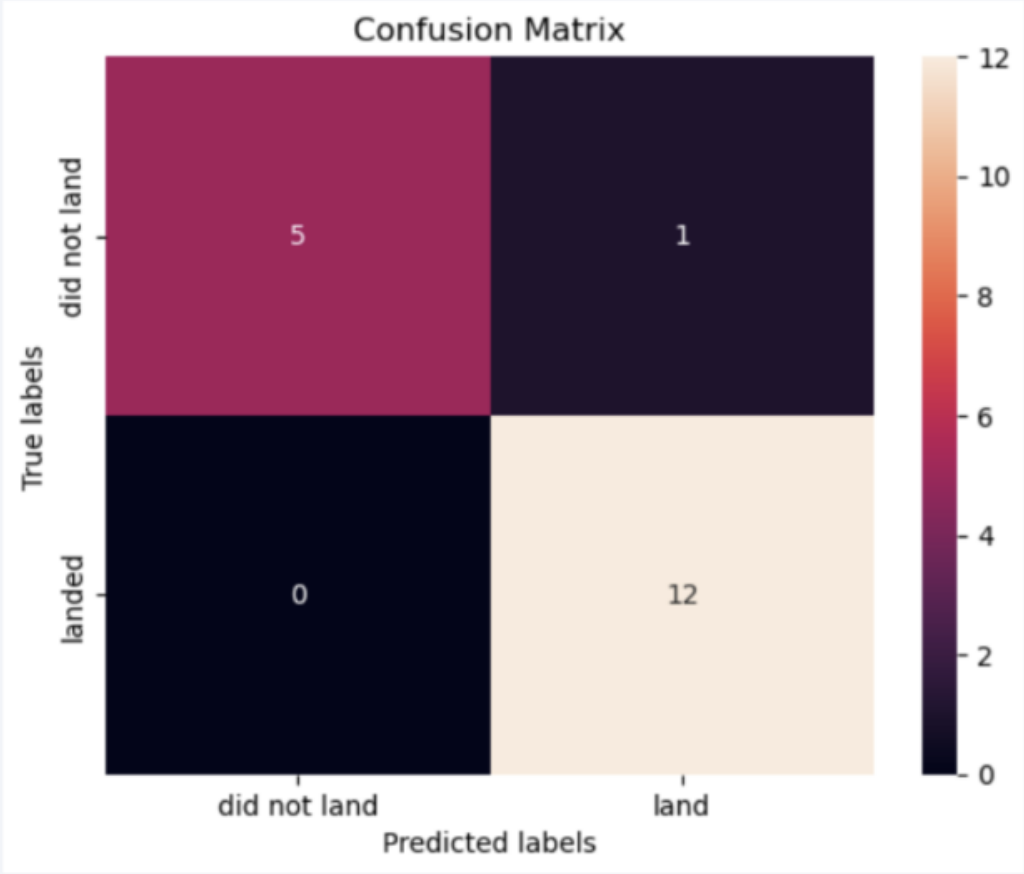
Classification Accuracy

- The Decision Tree model has the highest classification accuracy on the test data



Confusion Matrix

- The Decision Tree classifier model is almost entirely true negative and true positives, with no false negatives and only 1 false positive



Conclusions

- The KNN algorithm is the best model in machine learning for this dataset
- Launch site KSC LC-39A has the highest success rate
- Orbit GEO_HEO & ES-L1 have the highest success rate
- Launches that have a low payload have a higher success rate
- Success rate is increasingly directly proportional with time

Thank you!

